BAU – FET – DCE. EE- BSE0401 MIDTERM EXAM 45 MIN

- SHOW YOUR COMLETE CALCULATIONS AND FILL IN THE SPACES BELLOW.

Q1- The missing Factor in the equation (?)(A/G,i.n) = (A/F,i.n) is?

[The Factor (..G..../..F...,i.n.)]

Q2- What is the Interest Amount (I) after 6 years, for an original (first) amount of JD 1300, at a Simple Interest Rate i=10% per year?

[Interest Amount =780....]

Q3- How long will it takes to Withdraw JD 2100 from a deposit of JD 1100 at i=7% per year?

[n= 9.557.. years]

Q4- Given a project with an Operating Cost = 300, an Income at the end of the 4th year = 2400, and a S.V. = 1500. If i=7% per year and n=6 years, then the Equivalent Single Value at the end of the 4th year is?

[Eq. X (4) = ~~-2316,28~~ -1835,29 **.....]**

Q5- A project with an Annual Incomes for the first 4 years =1000, if i=7% per year and n=8 years, then the Equivalent Gradient Value (G) for the last 4 years is?

[Eq. G = -925,965 **.....]**

Q6- A project with an Extra Cost, at the end of the 3rd year = 1200, and an annual Income = JD 1000, Decreased by 100 yearly over n=6 years. If i=7% per year, then the equivalent Annual Value is around:

[Eq. A = -564,286 **.....]**

BEST WISHES.

Q1) $\left(\frac{G}{F}\right) * \left(\frac{A}{G}\right) = \frac{A}{F}$ ✓ #1

Q2) $I = P * i * n$

$= 1300 * 0.1 * 6$

$I = 780$ JD ✓ #2

Q3) $F = 2100$

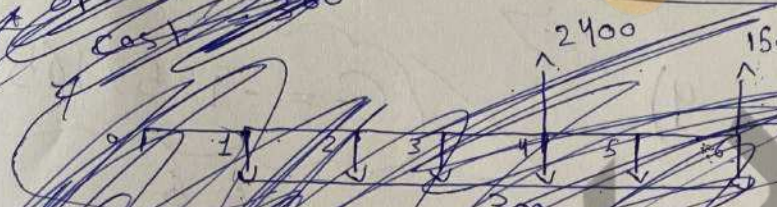
$P = 1100$

$i = 0.07$

$F_n = P * (1+i)^n \Rightarrow \ln(F_n/P) = n \cdot \ln(1+i)$

$n = \frac{\ln\left(\frac{2100}{1100}\right)}{\ln(1+0.07)} \Rightarrow n = 9.557$ ✓ #3

Q4)



$x(4) = -2400 - 1500(P/F, 7\%, 2) + 300(P/A, 7\%, 6)(F/P, 7\%, 4)$

$= -2400 - 1500(0.8734) + 300(4.767)(1.311)$

$x(4) = -1835.239$

Q4)



$x(4) = 300(F/P, 7\%, 4) - 1500(P/F, 7\%, 2) - 2400$

$300(1.311) - 1500(0.8734) - 2400$

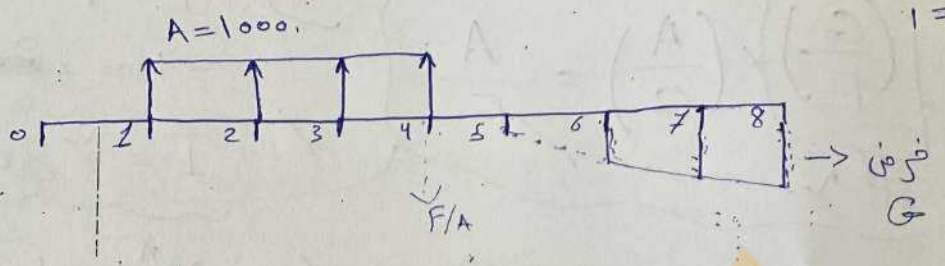
$x(4) = -3316.8$

$x(4) = -2400 - 1500(0.8734) + 300(4.767)(1.311)$

$x(4) = -1835.239$

Q5

$i = 7\%$



$$X(4) = -1000\,(F/A, 7\%, 4)$$

$$= -1000\,(4.44)$$

$$X(4) = -4440 = P$$

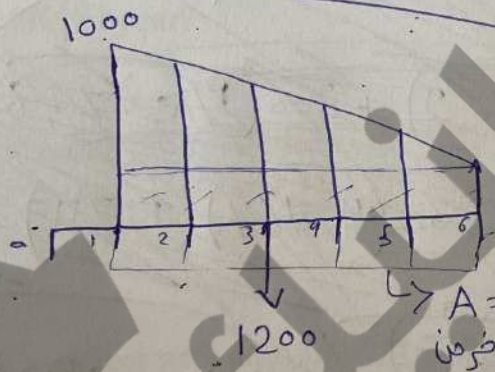
$$P = G\,(P/G, 7\%, 4)$$

$$G = \frac{-4440}{4.795}$$

$$G = -925.965$$

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Q6



$A = 1000$

$G = 100$

$$X(0) = 1200\,(P/F, 7\%, 3) - 1000\,(P/A, 7\%, 6) + 100\,(P/G, 7\%, 6)$$

$$= 1200\,(0.8163) - 1000\,(4.767) + 100\,(10.978)$$

$$X(0) = -2689.64 = P$$

$$A = P\,(A/P, 7\%, 6) = -2689.64\,(0.2098)$$

$$A = -564.286$$

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